

AdaRadar: Rate Adaptive Spectral Compression for Radar-based Perception

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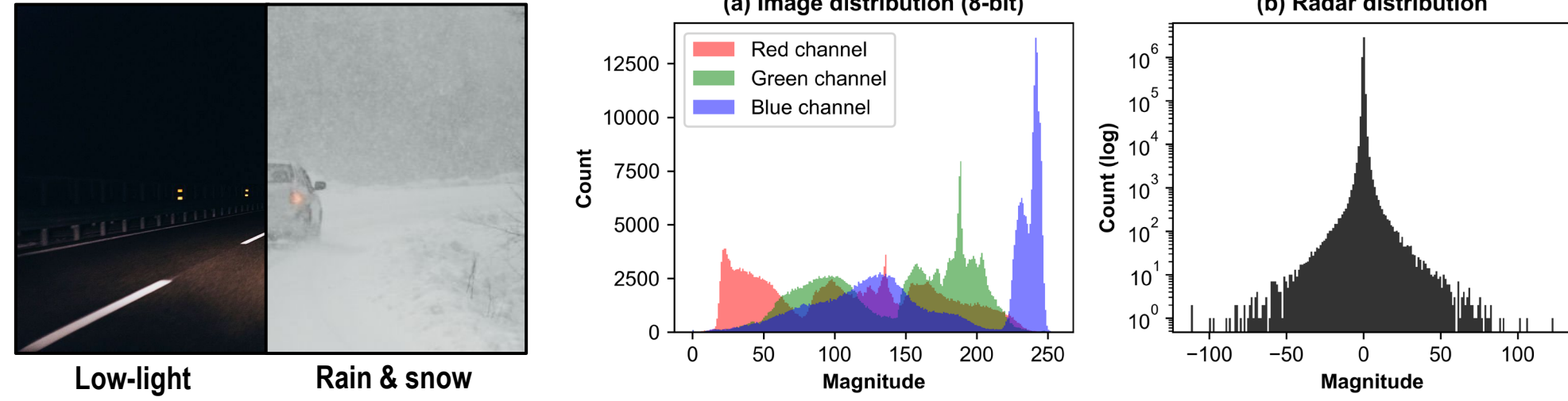


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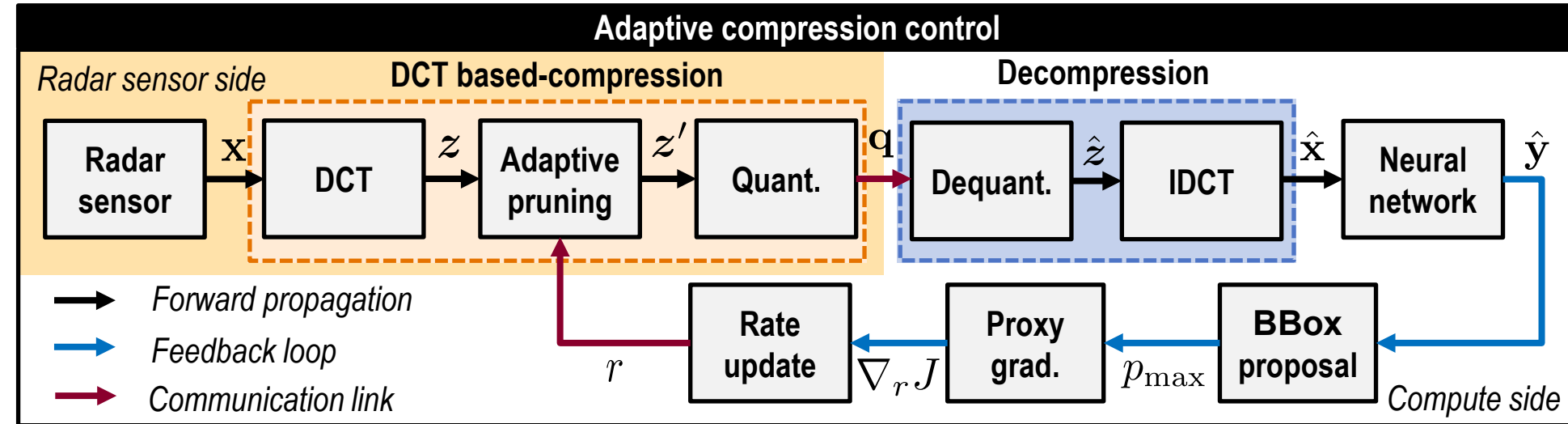
Project page

Radar Complements Camera & LiDAR



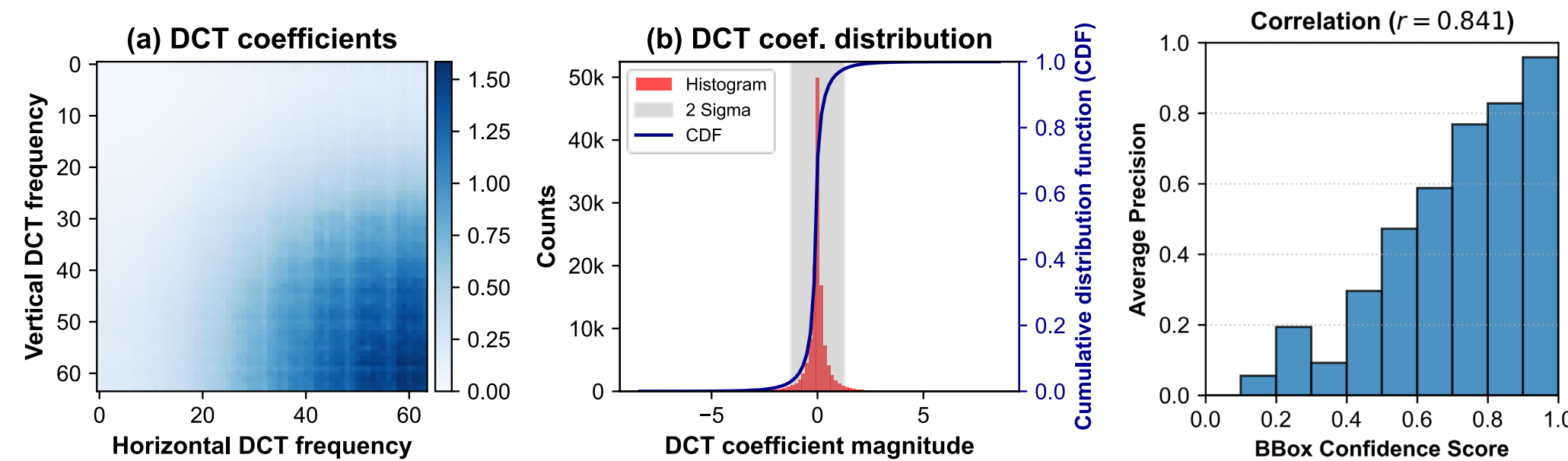
- **Radar perception** is a critical modality in autonomous driving, complementing camera & LiDAR.
- **Natural images** span a broad dynamic range.
- Radar signals exhibit extreme **sparsity** and a heavy-tailed profile (1st/99th percentiles at ± 1.74).

Adaptive Compression Rate Control



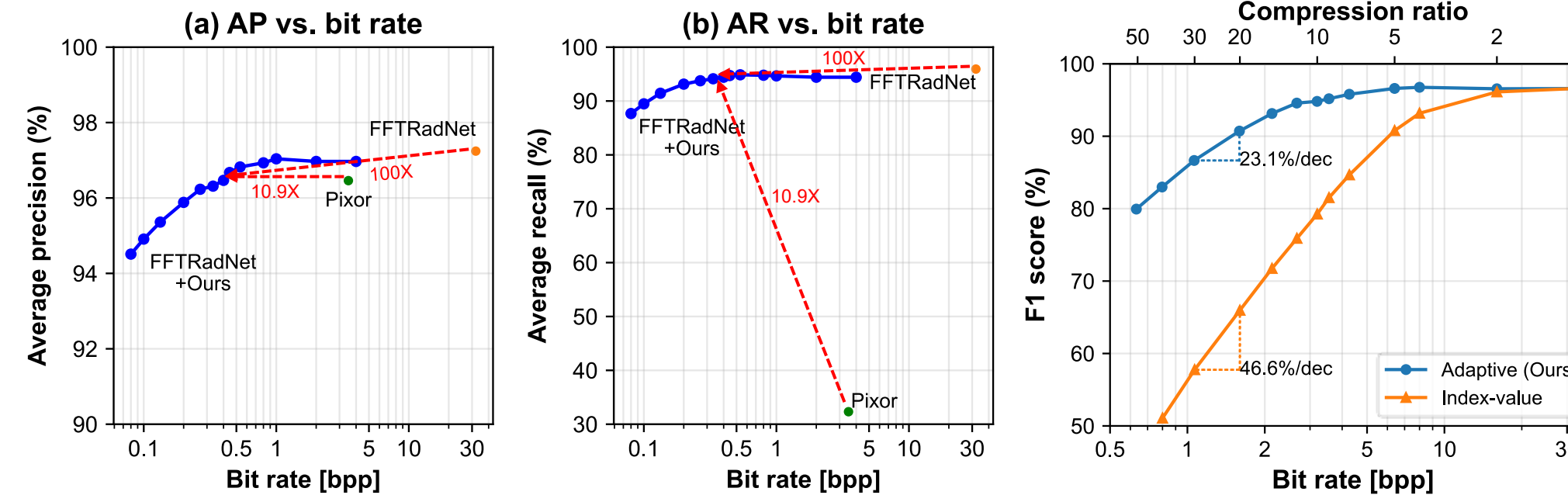
- Our proposed method introduces a **feedback loop** in which the proxy gradient is computed from the detection outputs to update the compression ratio adaptively.
- This avoids the need for **backpropagation** through the communication channel.
- DCT-based compression on the radar sensor side and decompression on the compute side.

Motivation for Compression & Adaptation



- The DCT coefficient **magnitudes** are clustered in the high-frequency bins.
- Their histogram is sharply peaked—strong **sparsity** and clear opportunities for compression.
- BBox confidence shows great correlation with the ground truth label.

Qualitative Results



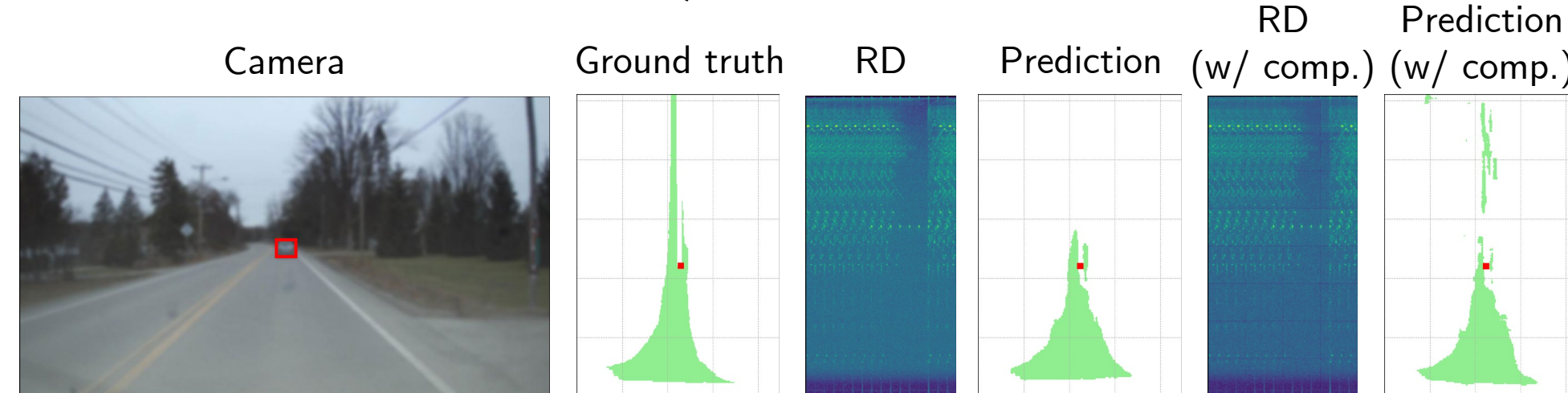
- Our compression scheme achieves **100x** reduction in the radar feature map size while having a **1%p decrease** in the performance compared to the baseline.
- Our method remains **stable** until 5x compression whereas the performance drop-off begins immediately with the index-value pair-based method.
- The roll-off gradient of **23.1%/dec** is much more gentle for the spectral pruning-based compression vs. **46.6%/dec** of the spatial-domain counterpart.

Compression Results – RADial Dataset

Method	Bit	Prune ratio	Bit rate [bpp] ↓ (Comp. ratio ↑)	P (%) ↑	Detection R (%) ↑	F1 (%) ↑	Segment. mIoU (%) ↑
Baseline [2]	32	-	32 (1x)	97.24	95.93	96.58	75.97
+Index-value [11]	32	12x	2.67 (12x)	97.55	62.12	75.91	49.86
+Adaptive (Ours)	4	12.57x	0.32 (101x)	96.25	94.04	95.13	79.34

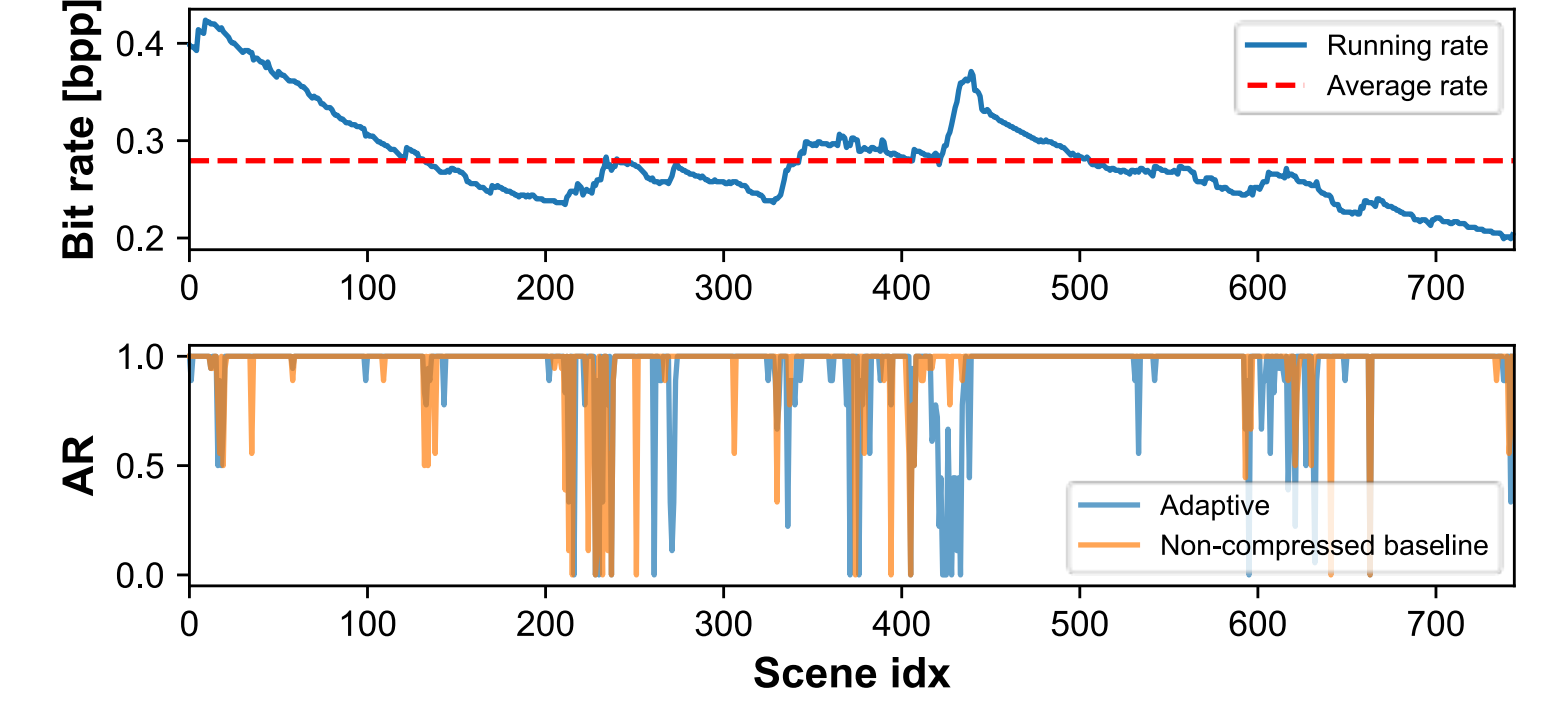
- An average pruning ratio of **12.57x** combined with **8x** gain from quantization, yields a bit rate of **0.32 bits per pixel (bpp)**, corresponding to a **101x** compression ratio.
- On the segmentation task, ours improves the performance by **4%**.
- Our method markedly **surpasses** index-value based method.

Qualitative Results



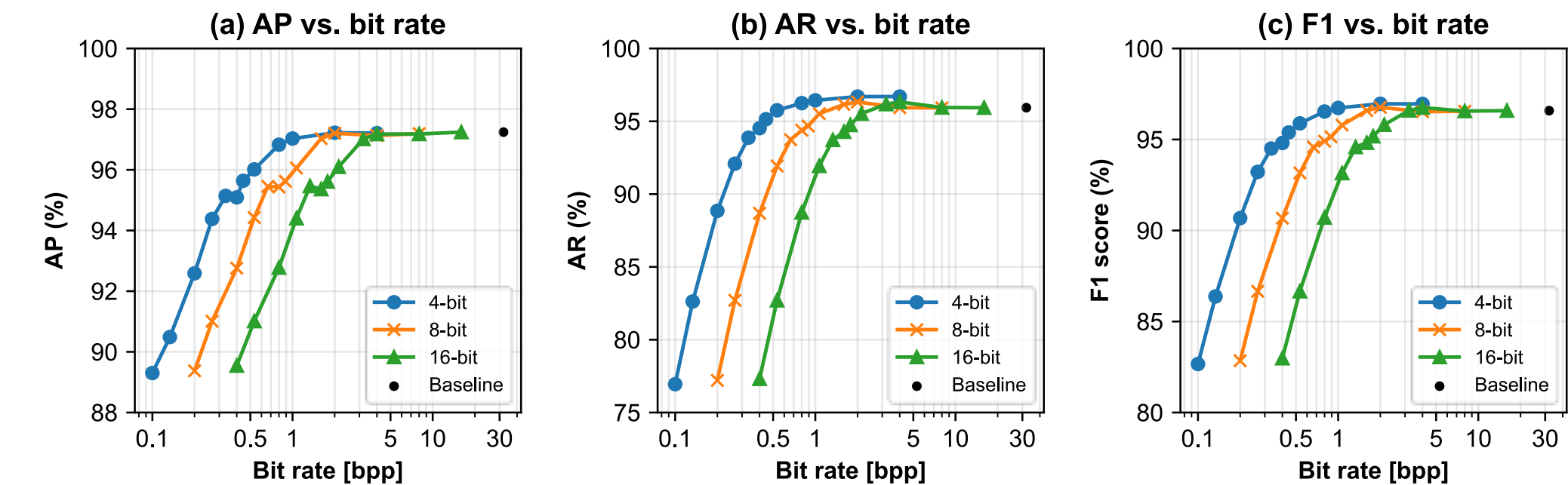
- **Camera images** provide contextual reference for the scene (not fed into network).
- The network predicts both **detection** and **segmentation** outputs.
- Compressed range-Doppler (RD) map leads to **improved performance** compared to the uncompressed baseline.

Time Series Analysis



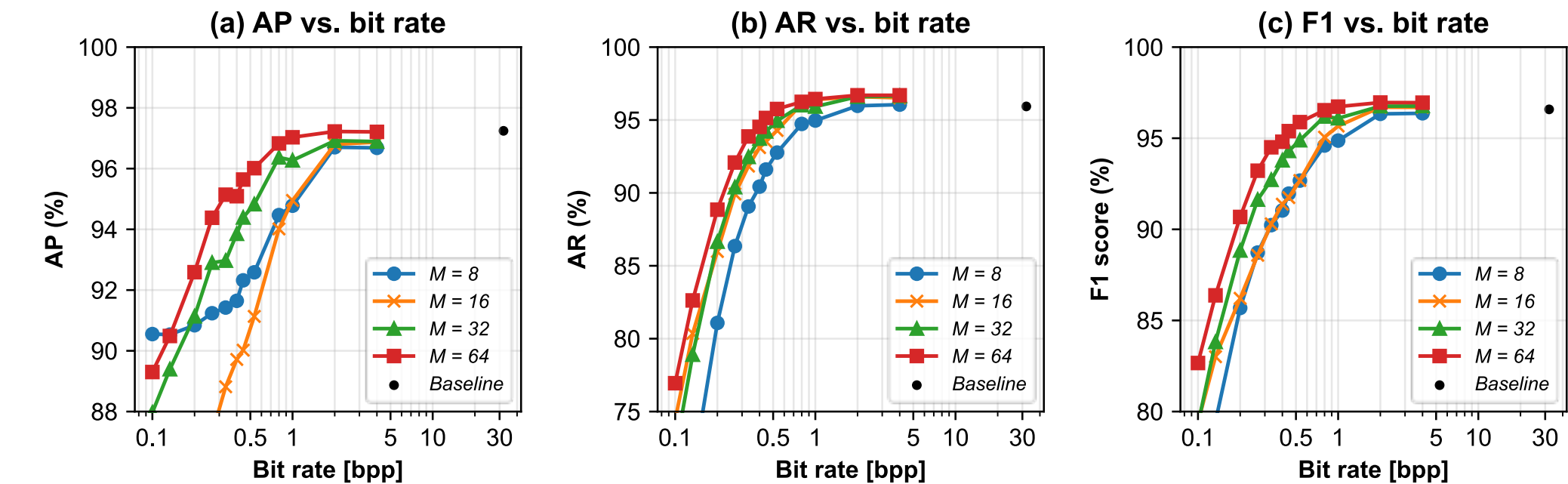
- We expect that the controller decreases the pruning ratio to compensate for the AR drop.
- It achieves an average bit rate of **0.279 bpp** with 8-bit quantization, yielding a **115x** compression with AR of **93.91%**.

Effect of Quantization Bit Width



- Quantization up to **4-bit** does not affect the performance compared to that of 8-bit and 16-bit. ($M = 64$)

Effect of DCT Block Length



- For the block-size sweep, **M = 64** yields the best trade-off between performance and overhead from the scaling factor. (bit width = 4-bit)